

Python Basics Assignment

Topic: Data Types, Operators, Lists, Tuples, Sets & Dictionaries

Time Allowed: 60 Minutes

Instructions for Students

- Read each question carefully before answering.
- Write clean, readable code with proper formatting.
- For coding tasks, ensure variable names are descriptive.

Part 1: Multiple Choice Questions (10 Marks)

Select the correct answer for each question. (2 Marks each)

1. What is the data type of `x = 5 / 2` in Python 3?
 - ☐ A) int
 - ☐ B) float
 - ☐ C) double
 - ☐ D) str
2. Which operator is used for exponentiation (power calculation)?
 - ☐ A) ^
 - ☐ B) **
 - ☐ C) %
 - ☐ D) //
3. What will be the output of `type((5,))`?
 - ☐ A) int
 - ☐ B) tuple
 - ☐ C) list
 - ☐ D) set
4. Which built-in data structure does **NOT** allow duplicate elements?
 - ☐ A) List
 - ☐ B) Tuple
 - ☐ C) Set
 - ☐ D) Dictionary values
5. How do you access the value associated with key 'age' in `person = {'name': 'Alice', 'age': 25}`?
 - ☐ A) `person[0]`

- B) person.age
- C) person['age']
- D) person.get(0)

Part 2: Code Output & Debugging (10 Marks)

Analyze the code snippets and answer accordingly. (2 Marks each)

1. **Floor Division & Modulo:** What will be printed by the following code?
2. `a = 17`
3. `b = 5`
4. `print(a // b, a % b)`
5. **List Slicing:** What is the output of this slice?
6. `numbers = [10, 20, 30, 40, 50, 60]`
7. `print(numbers[1:4])`
8. **Set Operations:** What will the following code output?
9. `set_a = {1, 2, 3, 4}`
10. `set_b = {3, 4, 5, 6}`
11. `print(set_a & set_b)`
12. **Tuple Immutability:** Explain what happens when this code is executed:
13. `tup = (1, 2, 3)`
14. `tup[0] = 10`
15. **Dictionary Properties:** What will `len(student)` print at the end of this script?
16. `student = {"name": "John", "score": 85}`
17. `student["score"] = 92`
18. `student["passed"] = True`
19. `print(len(student))`

Part 3: Practical Coding Challenges (30 Marks)

Problem 1: Temperature Converter (5 Marks)

Write a Python program that:

1. Prompts the user to enter a temperature in Celsius.
2. Converts the temperature to Fahrenheit using the formula: $F = (C \times \frac{9}{5}) + 32$.
3. Prints the converted temperature rounded to 2 decimal places.

Problem 2: List Manipulation & Statistics (7 Marks)

Given the list: grades = [78, 85, 62, 90, 85, 95, 78]

Write a Python script to:

1. Remove all duplicate values using a set and convert it back to a list.
2. Sort the unique grades in descending order.
3. Print the maximum, minimum, and average grade.

Problem 3: Developer Skill Sets (8 Marks)

Write a Python script to perform the following set operations:

1. Create a set named frontend_skills containing: 'HTML', 'CSS', 'JavaScript', 'React'.
2. Create a set named backend_skills containing: 'Python', 'SQL', 'JavaScript', 'Node'.
3. Calculate and display:
 - **All skills** known across both domains (Union).
 - **Shared skills** that exist in both domains (Intersection).
 - **Backend-only skills** (Difference: backend_skills - frontend_skills).

Problem 4: Student Database using Dictionaries (10 Marks)

Create a Python script that manages nested student records:

1. Initialize a dictionary named classroom with the following structure:
2. classroom = {
3. "S101": {"name": "Alice", "subjects": {"Math": 88, "Science": 92}},
4. "S102": {"name": "Bob", "subjects": {"Math": 75, "Science": 80}}
5. }
6. Add a new student record for "S103":
 - Name: "Charlie"
 - Marks: Math: 95, Science: 89
7. Calculate and print the average score for "Alice" across all her subjects.
8. Update "Bob"'s Math score to 82.
9. Print the updated classroom dictionary.

Instructions for Students

- Write complete, runnable Python code for each problem.
- Follow proper variable naming conventions and output formatting.

Problem 1: Basic Math & String Formatting (8 Marks)

Write a Python script that calculates the total bill for a restaurant visit:

1. Store the bill amount 125.50 and tax percentage 8% in variables.
2. Store a tip percentage 15% in a variable.
3. Calculate:
 - Tax amount: $\text{\text{\$}}\{\text{bill}\} \times \left(\frac{\text{\text{\$}}\{\text{tax}\}}{100}\right)$
 - Tip amount: $\text{\text{\$}}\{\text{bill}\} \times \left(\frac{\text{\text{\$}}\{\text{tip}\}}{100}\right)$
 - Grand total: $\text{\text{\$}}\{\text{bill}\} + \text{\text{\$}}\{\text{tax}\} + \text{\text{\$}}\{\text{tip}\}$
4. Print a formatted summary showing the original bill, tax, tip, and grand total, each rounded to 2 decimal places.

Problem 2: List Slicing & Operations (10 Marks)

Given a list of numbers:

numbers = [12, 45, 2, 91, 33, 24, 88, 50, 19, 73]

Write a Python script to perform the following:

1. Extract and print the sublist containing the middle 4 elements using slicing.
2. Append the number 100 to the end of numbers.
3. Insert the number 5 at index 0.
4. Sort the entire list in ascending order.
5. Print the updated list and its final length using len().

Problem 3: Dictionary & Set Analysis (12 Marks)

You are managing inventory for a small electronics shop.

1. Create a dictionary named inventory:
2. inventory = {
3. "laptop": 12,
4. "mouse": 50,
5. "keyboard": 25,
6. "monitor": 8
7. }
8. Perform the following operations:
 - Add a new item "headphones" with a quantity of 30.
 - Update the quantity of "mouse" to 45 after a sale.
 - Remove "monitor" from the dictionary.
9. Create a set in_stock_items containing all item names from the inventory dictionary.

10. Create a second set `discounted_items = {"mouse", "keyboard", "webcam"}`.

11. Print:

- Items that are both in stock **and** discounted (Intersection).
- Discounted items that are currently **out of stock** (Difference: `discounted_items - in_stock_items`).

Problem 4: Tuple Data Packing & Unpacking (10 Marks)

Write a Python script that works with employee records stored as tuples:

1. Create a tuple containing employee details: `employee = ("E402", "Sarah Connor", "Engineering", 75000)`.
2. Unpack the tuple into four separate variables: `emp_id`, `emp_name`, `department`, and `salary`.
3. Print a formatted string using these variables:

"Employee Sarah Connor (ID: E402) works in Engineering with a salary of \$75000."

4. Attempting to modify a tuple directly causes an error. Demonstrate how to update Sarah's salary to 82000 by converting the tuple into a list, making the edit, and converting it back to a tuple.

Problem 5: Nested Dictionary Aggregation (10 Marks)

A store tracks daily fruit sales using nested dictionaries:

```
sales = {  
    "Monday": {"apples": 10, "bananas": 15, "oranges": 8},  
    "Tuesday": {"apples": 12, "bananas": 20, "oranges": 5},  
    "Wednesday": {"apples": 8, "bananas": 18, "oranges": 12}  
}
```

Write a Python script to:

1. Calculate the **total number of apples** sold across all three days combined.
2. Calculate the **total number of bananas** sold across all three days combined.
3. Print which fruit sold more in total between apples and bananas.

Task 1: Type Errors & Input Conversion (10 Marks)

Buggy Code:

```
age = input("Enter your age: ")  
years_to_100 = 100 - age  
print("You will turn 100 in " + years_to_100 + " years!")
```

Questions:

1. Identify **two** distinct errors in the code snippet above. (4 Marks)
2. Explain why each error happens when running the code. (2 Marks)
3. Rewrite the code so it executes correctly. (4 Marks)

Task 2: Operator Precedence & Logic Flaws (10 Marks)

Buggy Code:

Goal: Calculate average score of 3 subjects and check if student passed (average >= 50)

math = 60

science = 40

english = 80

average = math + science + english / 3

is_passed = average >= 50 and math > 50 or science > 50 or english > 50

print("Average:", average)

print("Passed:", is_passed)

Questions:

1. Why does average calculate incorrectly in this code? (3 Marks)
2. Identify the issue with how is_passed evaluates boolean logic operator precedence. (3 Marks)
3. Provide the corrected Python script. (4 Marks)

Task 3: Mutability & Modifying Collections (10 Marks)

Buggy Code:

Goal: Store unique user tags and update user profile

tags = {"python", "coding", "python", "developer"}

user_profile = {

 "username": "coder123",

 "skills": tags,

 "location": "New York"

}

Line 9: Try to access skill by index

```
first_skill = tags[0]
```

```
# Line 12: Add new skill tag
```

```
tags.add("backend")
```

```
# Line 15: Try to update location using append
```

```
user_profile["location"].append("USA")
```

Questions:

1. What error occurs on **Line 9** and why? (3 Marks)
2. What error occurs on **Line 15** and why? (3 Marks)
3. Provide the corrected code that achieves all goals without errors. (4 Marks)

Task 4: Tuple and List Manipulation Errors (10 Marks)

Buggy Code:

```
coordinates = (10.5, 20.3, 30.1)
```

```
# Bug 1
```

```
coordinates[0] = 12.0
```

```
# Bug 2
```

```
numbers = [1, 2, 3, 4, 5]
```

```
numbers.remove(10)
```

```
# Bug 3
```

```
numbers.sort()
```

```
sorted_numbers = numbers.append(6)
```

```
print(sorted_numbers)
```

Questions:

1. Identify the bug in Bug 1 and explain tuple constraints. (3 Marks)
2. What error does Bug 2 cause and how do you prevent it? (3 Marks)
3. Why does sorted_numbers print None in Bug 3? Fix it. (4 Marks)

Task 5: Dictionary Access & Modification (10 Marks)

Buggy Code:

```
inventory = {  
    "apple": 50,  
    "banana": 30,  
    "orange": 20  
}  
  
# Goal: Safely retrieve "grape" quantity without crashing  
grape_qty = inventory["grape"]  
  
# Goal: Add 10 to "banana" quantity  
inventory.banana = inventory.banana + 10  
  
# Goal: Print dictionary keys as a tuple  
keys_tuple = tuple(inventory.keys)  
  
print(grape_qty, inventory, keys_tuple)
```

Questions:

1. Explain the three errors present in this snippet. (6 Marks)
2. Write the corrected code snippet that fulfills all three goals safely. (4 Marks)